

Module Interaction & Implementation Overview

5.8 Module Interaction

ComicCraft follows a modular architecture where the User Interface sends input to the FastAPI backend. The backend coordinates the Story Generation Module (Gemini Flash & Gemini Pro), the Image Generation Module (Stable Diffusion), the Layout Builder, and finally the PDF Export Module to generate the downloadable comic.

Workflow

User → User Interface → FastAPI Backend → Story Generation → Image Generation → Layout Builder → Comic Preview → PDF Export → Download PDF

Advantages

- Modular design.
- Easier maintenance.
- Independent testing.
- Better scalability.

6.1 Implementation Overview

The application integrates HTML, CSS, Jinja2, FastAPI, Gemini Flash, Gemini Pro, Stable Diffusion, and FPDF into a unified workflow. User inputs are processed by the backend, AI models generate stories and illustrations, the Layout Builder assembles comic panels, and the PDF Export Module produces a professional downloadable comic.

Benefits

- Fully automated workflow.
- High-quality AI-generated comics.
- Responsive interface.
- Efficient module communication.
- Scalable architecture.